Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If the area of the triangle on the complex plane formed by the points z, iz and $z + iz$ is 200 square units, then $|z|$ is

(A) 5

(B) 20

(C) 15

(D) none of these

Q.2

If $k + |k + z^2| = |z|^2$ where k is real, then a possible argument of z is

(A) 0

(B) π

(C) $\pi/2$

(D) none of these

Q.3

Let $z = a\left(\cos\frac{\pi}{5} + i\sin\frac{\pi}{5}\right), a \in \mathbf{R}, |a| < 1$, then $S = z^{2015} + z^{2016} + z^{2017} + \ldots$ equals

(A) $\frac{a^{2015}}{z-1}$

(B) $\frac{a^{2015}}{1-z}$

(C) $\frac{z^{2015}}{1-a}$

(D) $\frac{z^{2015}}{a-1}$

Q.4

Number of ordered pairs (a, b) of real numbers such that $(a + ib)^{2020} = a - ib$

(A) 2020

(B) 2021

(C) 2019

(D) 2022

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Q.5

If α and β are the roots of the equation $u^2 - 2u + 2 = 0$ and if $\cot\theta = x + 1$, then $\frac{(x+\alpha)^n - (x+\beta)^n}{\alpha - \beta}$ is equal to:

(A) $\frac{\sin n\theta}{\sin^n \theta}$

(B) $\frac{\cos n\theta}{\cos^n \theta}$

(C) $\frac{\sin n\theta}{\cos^n \theta}$

(D) $\frac{\cos n\theta}{\sin^n \theta}$

Q.6

If $\log_{1/2} \frac{|z|^2+2|z|+4}{2|z|^2+1} < 0$

(A) $|z| < 3$

(B) $1 < |z| < 3$

(C) $|z| > 1$

(D) $|z| < 2$

Q.7

Let z_1, z_2 be two complex numbers represented by points on the circle $|z| = 1$ and $|z| = 2$ respectively then

(A) $\max . |2z_1 + z_2| = 4$

(B) min. $|z_1 - z_2| = 1$

(C) $|z_2 + 1/z_1| \leq 3$

(D) all of these

Q.8

For any complex number z, minimum value of $|z| + |z - 1|$ is equal to

(A) 0

(B) $\frac{1}{2}$

(C) 1

(D) $\frac{3}{2}$

Q.9

If $\left|\frac{z_1 - 3z_2}{3 - z_1\bar{z}_2}\right| = 1$ and $|z_1| \neq 3$ then $|z_2|$ is

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(A) 3

(B) 1

(C) 2

(D) 4

Q.10

If $z = i^{i^i}$ then $z =$

(A) $-i$

(B) $e^{\frac{-\pi}{2}}$

(C) $e^{\frac{\pi}{2}}$

(D) none of these

Q.11

If $(\cos\theta + i\sin\theta)(\cos 2\theta + i\sin 2\theta)\ldots(\cos n\theta + i\sin n\theta) = 1$ then the value of θ is, $m \in Z$

(A) $4m\pi$

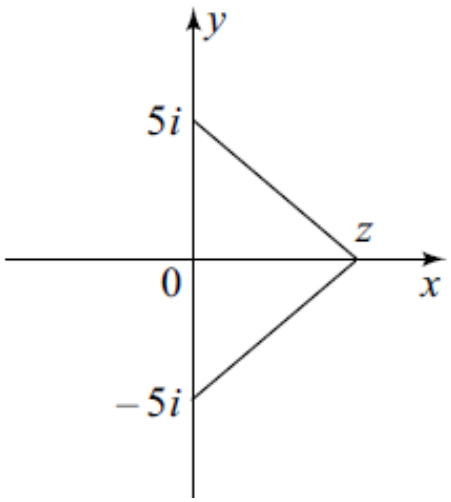
(B) $\frac{2m\pi}{n(n+1)}$

(C) $\frac{4m\pi}{n(n+1)}$

(D) $\frac{m\pi}{n(n+1)}$

Q.12

The complex numbers $z = x + iy$ which satisfy the equation $\left|\frac{z-5i}{z+5i}\right| = 1$, lie on



(A) the $x-$ axis

(B) the straight line $y = 5$

(C) a circle passing through origin

(D) none of these

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Q.13

If z_1 and z_2 are two non-zero complex numbers such that $|z_1+z_2|^2=|z_1|^2+|z_2|^2$, then

(A) $z_1\bar{z}_2$ is purely real

(B) $\frac{z_1}{z_2}$ is purely real

(C) $z_1\bar{z}_2+\bar{z}_1z_2=1$

(D) O, z_1, z_2 are the vertices of a right triangle

Q.14

One of the values of $(-i)^{1/3}$ is equal to

(A) $\frac{1+i\sqrt{3}}{2}$

(B) $\frac{1-i\sqrt{3}}{2}$

(C) $\frac{-\sqrt{3}-i}{2}$

(D) None of these

Q.15

If $|z|=1$ and $\omega=\frac{z-1}{z+1}(z\neq 1)$, then $\mathrm{Re}(\omega)$ is

(A) 0

(B) 1

(C) $\sqrt{2}$

(D) $\frac{\sqrt{2}}{|z+1|^2}$

Q.16

If $\left|z-\frac{4}{z}\right|=2$, then the greatest value of $|z|$ is

(A) $\sqrt{2}+1$

(B) $\sqrt{3}+1$

(C) $\sqrt{5}+1$

(D) none of these

Q.17

If z is a non-real complex number, then the minimum value of $\frac{\mathrm{Im}\, z^5}{(\mathrm{Im}\, z)^5}$ is:

(A) -1

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(B) -2

(C) -4

(D) -5

Q.18

If z_1, z_2, z_3 are three distinct complex numbers and a, b, c are three positive real numbers such that $\frac{a}{|z_2-z_3|} = \frac{b}{|z_3-z_1|} = \frac{c}{|z_1-z_2|}$, then value of $\frac{a^2}{z_2-z_3} + \frac{b^2}{z_3-z_1} + \frac{c^2}{z_1-z_2}$ is
(where ω, ω^2 are complex cube root of unity)

(A) 0

(B) ω

(C) ω^2

(D) none of these

Q.19

The real part of $\frac{1}{1-\cos\theta+i\sin\theta}$ is

(A) $\frac{1}{1-\cos\theta}$

(B) $\frac{1}{2}$

(C) $\frac{\tan\theta}{2}$

(D) 2

Q.20

If $|z-1| = |z+1| = |z-2i|$, then value of $|z|$ is

(A) 1

(B) 2

(C) $5/4$

(D) $3/4$

Q.21

If $b+ic = (1+a)z$ and $a^2+b^2+c^2 = 1; a, b, c \in R$, then $\frac{1+iz}{1-iz}$ is equal to $\frac{a+ib}{n+c}$ where n is ____.

Q.22

If $z^2+z+1 = 0$, where z is a complex number, then the value of $\left(z+\frac{1}{z}\right)^2 + \left(z^2+\frac{1}{z^2}\right)^2 + \left(z^3+\frac{1}{z^3}\right)^2 + \ldots.. + \left(z^6+\frac{1}{z^6}\right)^2$

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Q.23

The modulus of $\sum_{k=1}^{6}\left[\sin\frac{2k\pi}{7}-i\cos\frac{2\pi k}{7}\right]$ is _____.

Q.24

Consider the equation: $x^4+x^2+1=0$. If x_1, x_2, x_3, x_4 are roots of this equation, then the value of $x_1^6+x_2^6+x_3^6+x_4^6$ is equal to

Q.25

One of the values of $\frac{\sqrt{5+12i}+\sqrt{5-12i}}{\sqrt{5+12i}-\sqrt{5-12i}}$ is equal to pi/q reduced to the lowest form where $pq>0$ and Then $p+q=$ ____.

Q.26

$\text{If } \omega=\left(\frac{z-i}{1+iz}\right)^n$,n integral, then the number of values of n in the range $[1,20]$ such that ω lies on the unit circle is ____.

Q.27

Let ω is an imaginary cube roots of unity then the sum of the digits of value of $2(\omega+1)\left(\omega^2+1\right)+3(2\omega+1)\left(2\omega^2+1\right)+\ldots\ldots+(n+1)(n\omega+1)\left(n\omega^2+1\right)$ for $\mathrm{n}=2$ is

Q.28

If $z_1, z_2, \ldots\ldots, z_n$ lie on the circle $|z|=2$, then value of $E=|z_1+z_2+\ldots\ldots+z_n|-4\left|\frac{1}{z_1}+\frac{1}{z_2}+\ldots\ldots+\frac{1}{z_n}\right|$ is

Q.29

If $x^2-2x\cos\theta+1=0$, then the value of $x^{2n}-2x^n\cos n\theta+1$ is equal to

Q.30

The greatest integer less than or equal to the modulus of sum $\sum_{n=1}^{13}\left(i^n+i^{n+1}\right)$, where $i=\sqrt{-1}$, equals ____.

Answers & Solutions

Q.1 Answer:

20

Solution:

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The triangle formed is an isosceles right angled triangle

Therefore it has an area

$\Rightarrow \frac{1}{2}|\mathrm{z}|^2$

$= 200$

$\Rightarrow |\mathrm{z}|^2 = 400$

$\Rightarrow |\mathrm{z}| = 20$

Hence

$\Rightarrow |\mathrm{z}| = 20$ **units.**

Q.2 Answer:

$\pi/2$

Solution:

$k + |k + z^2| = |z|^2$

Hence k **and** $\left|\mathrm{z}^2\right|$ **are collinear.**

Now $k \in R^-$

Since, k **and** z^2 **is collinear, then** z^2 **has to be purely real.**

Thus, $\arg\left(\mathrm{z}^2\right) = (2\mathrm{n} - 1)\pi$

Now $z = re^{i\theta}$

$\mathrm{z}^2 = \mathrm{r}^2\mathrm{e}^{2i\theta} = \mathrm{me}^{2i\theta}$

Hence $\arg\left(\mathrm{z}^2\right) = 2\arg(\mathrm{z})$

Therefore, $\arg(\mathrm{z}) = \frac{1}{2}\arg\left(\mathrm{z}^2\right)$

$= \frac{(2\mathrm{n}-1)\pi}{2}$

Putting $\mathrm{n} = 1$**, we get**

$\arg(\mathrm{z}) = \frac{\pi}{2}$

Hence, option 'C' is correct.

Q.3 Answer:

$\frac{a^{2015}}{z-1}$

Solution:

We have $|z| = |a| < 1$**, thus**

$S = \frac{z^{2015}}{1-z}$

But $z^{2015} = a^{2015}[\cos(403\pi) + i\sin(403\pi)] = -a^{2015}$

$\therefore S = \frac{a^{2015}}{z-1}$

Q.4 Answer:

2022

Solution:

If $(a, b) = (0, 0)$ **then the equation holds.**

Now, say $(a, b) \neq (0, 0)$ **and let** $z = a + ib$**, then we want:**

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$z^{2020} = \bar{z} = \frac{|z|^2}{z}$

$\Rightarrow z^{2021} = |z|^2 \Rightarrow$ **Total number of solutions is 2022 including** $(0, 0)$.

Q.5 Answer:

$\frac{\sin n\theta}{\sin^n \theta}$

Solution:

$u^2 - 2u + 2 = 0$

$\Rightarrow \quad \mathbf{u = 1 \pm i}$

So, $\alpha = 1 + \mathbf{i}$ **and** $\beta = 1 - \mathbf{i}$

Now given that, $x = \cot\theta - 1$

so, $\frac{(x+\alpha)^n - (x+\beta)^n}{\alpha - \beta} = \frac{(\cot\theta - 1 + 1 + i)^n - (\cot\theta - 1 + 1 - i)^n}{2i}$

$= \frac{(\cot\theta + i)^n - (\cot\theta - i)^n}{2i} = \frac{(\cos\theta + i\sin\theta)^n - (\cos\theta - \sin\theta)^n}{(\sin\theta)^n (2i)}$

$= \frac{e^{(in0)} - e^{-(in\theta)}}{(\sin\theta)^n 2i}$

Q.6 Answer:

$|z| < 3$

Solution:

$\log_{1/2} \frac{|z|^2 + 2|z| + 4}{2|z|^2 + 1} < 0 \quad \Rightarrow \frac{|z|^2 + 2|z| + 4}{2|z|^2 + 1} > 1$

or $|z|^2 + 2|z| + 4 > 2|z|^2 + 1$ **or** $|z|^2 - 2|z| - 3 < 0$

or $\quad (|z| + 1)(|z| - 3) < 0$

$\therefore |z| - 3 < 0.$

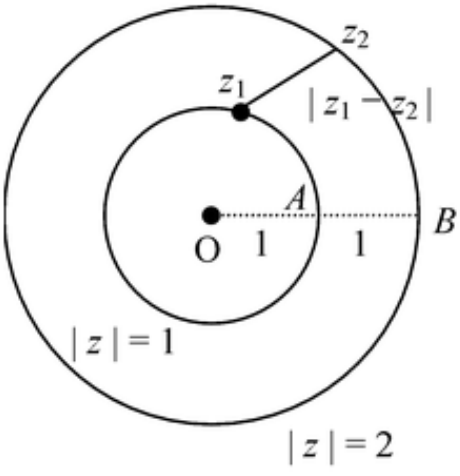
Q.7 Answer:

all of these

Solution:

$|2z_1 + z_2| \le |2z_1| + |z_2| = 2|z_1| + |z_2| = 2 \times 1 + 2 = 4$

From the figure, $|z_1 - z_2|$ **is the least when** $0, z_1, z_2$ **are collinear.**



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Then $|z_1 - z_2| = 1$

Again, $\left|z_2 + \frac{1}{z_1}\right| \leq |z_2| + \left|\frac{1}{z_1}\right| = 2 + \frac{1}{|z_1|} = 2 + \frac{1}{1} = 3$

Q.8 Answer:

1

Solution:

The triangle inequality tells that:

$|z| + |z-1| = |z| + |-(z-1)| \geq |z-(z-1)| = |z - z + 1| = |1| = 1$

$\implies |z| + |z-1| \geq 1$

So, the minimum value of $|z| + |z-1|$ **is 1**

Q.9 Answer:

1

Solution:

$\left|\frac{Z_1 - 3Z_2}{3 - Z_1\overline{Z_2}}\right| = 1$

$\Rightarrow |Z_1 - 3Z_2| = \left|3 - Z_1\overline{Z_2}\right|$

$\Rightarrow |Z_1 - 3Z_2|^2 = \left|3 - Z_1\overline{Z_2}\right|^2$

$\Rightarrow (Z_1 - 3Z_2)\left(\overline{Z_1} - 3\overline{Z_2}\right) = \left(3 - Z_1\overline{Z_2}\right)\left(3 - \overline{Z_1}Z_2\right)$

$\Rightarrow |Z_1|^2 - 3Z_1\overline{Z_2} - 3Z_2\overline{Z_1} + 9|Z_2|^2 = 9 - 3\overline{Z_1}Z_2 - 3Z_1\overline{Z_2} + |Z_1|^2|Z_2|^2$

$\Rightarrow |Z_1|^2 - |Z_1|^2|Z_2|^2 - 9 + 9|Z_2|^2 = 0$

$\Rightarrow |Z_1|^2\left(1 - |Z_2|^2\right) - 9\left(1 - |Z_2|^2\right) = 0$

$\Rightarrow \left(1 - |Z_2|^2\right)\left(|Z_1|^2 - 9\right) = 0$

$\Rightarrow |Z_2| = 1 \quad (\because |Z_1| \neq 3$ **given)**

Q.10 Answer:

$-i$

Solution:

$\mathbf{Z} = (i)^{(i)^{(i)}}$

$\mathbf{i} = \mathrm{e}^{\mathrm{i}\pi/2}$

$\mathrm{i}^{\mathrm{i}} = \mathrm{e}^{\mathrm{i}\cdot(\mathrm{i}\pi/2)} = \mathrm{e}^{-\pi/2}$

$\mathrm{i}^{\mathrm{i}} = \mathrm{e}^{\mathrm{i}(-\pi/2)}$

$\Rightarrow \cos(\pi/2) - \mathrm{i}\sin(\pi/2)$

$\Rightarrow 0 - \mathrm{i} = -\mathrm{i}$

Q.11 Answer:

$\frac{4m\pi}{n(n+1)}$

Solution:

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Writing in euler's form, we get

$e^{i\theta}e^{2i\theta}e^{3i\theta}\ldots e^{in\theta}=1$

$e^{i(\theta+2\theta+3\theta+\ldots n\theta}=1$

$e^{i\frac{n(n+1)}{2}}=e^{2\,m\pi}$

Therefore

$\frac{n(n+1)}{2}\theta=2\,m\pi$

$\theta=\frac{4m\pi}{n(n+1)}$

Q.12 Answer:

the $x-$ axis

Solution:

$\left|\frac{z-5i}{z+5i}\right|=1\Rightarrow|z-5i|=|z-(-5i)|$

$\Rightarrow$ **z lies on the perpendicular bisector of the segment joining $5i$ and $-5i$, i.e., z lies on the x-axis.**

Q.13 Answer:

O, z_1, z_2 are the vertices of a right triangle

Solution:

$|z_1+z_2|^2=|z_1|^2+|z_2|^2\Rightarrow(z_1+z_2)(\bar{z}_1+\bar{z}_2)=|z_1|^2+|z_2|^2$

$\Rightarrow\quad |z_1|^2+z_2\bar{z}_1+z_1\bar{z}_2+|z_2|^2=|z_1|^2+|z_2|^2$

$\Rightarrow\quad z_2\bar{z}_1+z_1\bar{z}_2=0\quad\Rightarrow\overline{z_1\bar{z}_2}=\bar{z}_1z_2=-z_1\bar{z}_2$

$\Rightarrow$ **$z_1\bar{z}_2$ is purely imaginary.**

$\Rightarrow$ **$\frac{z_1}{z_2}|z_2|^2$ is purely imaginary.**

Since $|z_2|^2$ is real, we get $\frac{z_1}{z_2}$ is purely imaginary.

As $\frac{z_1}{z_2}$ is purely imaginary, $\arg\left(\frac{z_1}{z_2}\right)=\frac{\pi}{2}$.

Thus O, z_1, z_2 forms a right triangle.

Q.14 Answer:

$\frac{-\sqrt{3}-i}{2}$

Solution:

$(-i)^{1/3}=\left(e^{-i\pi/2}\right)^{1/3}=\left(e^{-i\pi/2}\cdot e^{i2n\pi}\right)^{1/3}=e^{(i\pi/6)(4n-1)}=x_n$**, say where $n=0,1,2$.**

$x_0=\frac{\sqrt{3}-i}{2}$ **and** $x_1=-\left(\frac{\sqrt{3}+i}{2}\right)$

Q.15 Answer:

0

Solution:

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Given,

$|z| = 1 \quad \& \quad w = \frac{(z-1)}{(z+1)}$

To find $\mathrm{Re(w)}$

solution,

Let, $\mathrm{z = a + ib} \quad \therefore \mathrm{\overline{z} = a - ib}$

$|z| = \sqrt{a^2 + b^2} \quad |\bar{z}| = \sqrt{a^2 + b^2}$

We know that

$\mathrm{Re(w) = \frac{w+\overline{w}}{2}}$

$\mathrm{Re(w) = \frac{\frac{(z-1)}{(z+1)} + \frac{\overline{z}-1}{\overline{z}+1}}{2}}$

$\mathrm{Re(w) = \frac{(z-1)(\overline{z}+1)+(\overline{z}-1)(z+1)}{2(z+1)(\overline{z}+1)}}$

$\mathrm{Re(w) = \frac{z\overline{z}-\overline{z}+z-1+z\overline{z}+\overline{z}-z-1}{2(z+1)(\overline{z}+1)}}$

$\mathrm{Re(w) = \frac{2z\overline{z}-2}{2(z+1)(\overline{z}+1)}}$

$\mathrm{Re(w) = \frac{z\overline{z}-1}{(z+1)(\overline{z}+1)}}$

$\mathrm{Re}(w) = \frac{a^2+b^2-1}{(z+1)(\bar{z}+1)}$

Given, $a^2 + b^2 = 1$

Hence, $\mathrm{Re}(w) = 0$

Q.16 Answer:

$\sqrt{5} + 1$

Solution:

We have $|z| = \left|z - \frac{4}{z} + \frac{4}{z}\right|$

$\leq \left|z - \frac{4}{z}\right| + \frac{4}{|z|} = 2 + \frac{4}{|z|}$

$\Rightarrow \quad |z|^2 \leq 2|z| + 4 \Rightarrow (|z| - 1)^2 \leq 5 \Rightarrow |z| - 1 \leq \sqrt{5} \Rightarrow |z| \leq \sqrt{5} + 1$

$\therefore \quad$ **greatest value of** $|z| = \sqrt{5} + 1$

Q.17 Answer:

-4

Solution:

z is a non-real complex number.

$\mathrm{let\, z = re^{i\theta}}$

$\frac{\mathrm{Im}\, z^5}{(\mathrm{Im}\, z)^5} = \frac{r^5 \sin 5\theta}{r^5 \sin^5 \theta}$

say, $f(\theta) = \frac{\sin 5\theta}{\sin^5 \theta}$

equating first derivative to 0

$\mathrm{f'(\theta) = 0}$

$\Rightarrow \sin^5 \theta \cdot 5 \cos 5\theta - \sin 5\theta \cdot 5 \sin^4 \theta \cdot \cos \theta$

$\Rightarrow \sin \theta \cdot \cos 5\theta = \cos \theta \cdot \sin 5\theta$

$\Rightarrow \tan \theta = \tan 5\theta$

$\Rightarrow \theta = 0, \pi/4 \ldots$

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but $f(0)$ is indeterminate

$f(\pi/4) = -4$

$\therefore$ Minimum value of $\frac{\text{Im}z^5}{(\text{Im}z)^5}$ is -4

Q.18 Answer:

0

Solution:

Let $\frac{a}{|z_2 - z_3|} = \frac{b}{|z_3 - z_1|} = \frac{c}{|z_1 - z_2|} = k(\text{say})$

$\Rightarrow \quad a = k|z_2 - z_3|, b = k|z_3 - z_1|, c = k|z_1 - z_2|$

Now, $a^2 = k^2|z_2 - z_3|^2 = k^2(z_2 - z_3)(\bar{z}_2 - \bar{z}_3)$

$\Rightarrow \quad \frac{a^2}{z_2 - z_3} = k^2(\bar{z}_2 - \bar{z}_3)$

Thus, $\frac{a^2}{z_2 - z_3} + \frac{b^2}{z_3 - z_1} + \frac{c^2}{z_1 - z_2}$

$= k^2(\bar{z}_2 - \bar{z}_3) + k^2(\bar{z}_3 - \bar{z}_1) + k^2(\bar{z}_1 - \bar{z}_2) = 0$

Q.19 Answer:

$\frac{1}{2}$

Solution:

Given that: $(1 - \cos\theta + \text{i}\sin\theta)^{-1} = \frac{1}{1 - \cos\theta + \text{i}\sin\theta}$

$= \frac{1}{2\sin^2\frac{\theta}{2} + \text{i}\cdot 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}} = \frac{1}{2\sin\frac{\theta}{2}} \cdot \frac{1}{\sin\frac{\theta}{2} + \text{i}\cos\frac{\theta}{2}}$

$= \frac{1}{2\sin\frac{\theta}{2}} \cdot \frac{1}{\sin\frac{\theta}{2} + \text{i}\cos\frac{\theta}{2}} \cdot \frac{\sin\frac{\theta}{2} - \text{i}\cos\frac{\theta}{2}}{\sin\frac{\theta}{2} - \text{i}\cos\frac{\theta}{2}}$, rationalize denominator

$= \frac{1}{2\sin\frac{\theta}{2}}\left(\sin\frac{\theta}{2} - \text{i}\cos\frac{\theta}{2}\right)$

$= \frac{1}{2} - \frac{\text{i}}{2}\cot\frac{\theta}{2}$

Q.20 Answer:

$3/4$

Solution:

z is centre of the circle passing through $1 + 0i$, $-1 + 0i$ and $0 + 2i$. Clearly centre lies on the y-axis. If $z = 0 + ai$ the centre, then

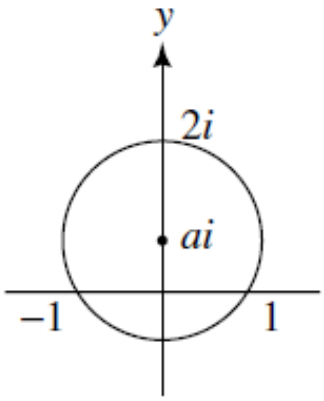
$\sqrt{1 + a^2} = |a + 2|$

$\Rightarrow \quad 1 + a^2 = a^2 + 4a + 4$

$\Rightarrow a = 3/4$

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$\therefore |z| = 3/4$



Q.21 Answer:

—

Solution:

$iz = \frac{-c+ib}{1+a} \Rightarrow \frac{1+iz}{1-iz} = \frac{1+a-c+ib}{1+a+c-ib} \times \frac{[(1+a+c)+ib]}{[(1+a+c)+ib]}$
$= \frac{(1+a+ib)^2-c^2}{2+2(a+c+ac)} = \frac{(1+a)(a+ib)}{(1+a)(1+c)} = \frac{a+ib}{1+c}$

Q.22 Answer:

—

Solution:

$z^2+z+1=0 \quad \Rightarrow \quad z=\omega,\omega^2$
$\Rightarrow \left(z+\frac{1}{z}\right)^2+\left(z^2+\frac{1}{z^2}\right)^2+\left(z^3+\frac{1}{z^3}\right)^2+\left(z^4+\frac{1}{z^4}\right)^2+\left(z^5+\frac{1}{z^5}\right)^2+\left(z^6+\frac{1}{z^6}\right)^2$
$=1+1+4+1+1+4=12$

Q.23 Answer:

—

Solution:

We have $\sum_{k=1}^{6}\left[\sin\frac{2k\pi}{7}-i\cos\frac{2k\pi}{7}\right] = -i\sum_{k=1}^{6}e^{i\cdot\frac{2\pi k}{7}} = -i\sum_{k=1}^{6}\left(e^{i\cdot\frac{2\pi}{7}}\right)k$
Let $e^{i\frac{2\pi}{7}}=\alpha \Rightarrow \quad -i\sum_{k=1}^{6}\alpha^k = -i\left[\alpha+\alpha^2\ldots\ldots\alpha^6\right]$
$\left[\frac{-i\alpha\left(1-\alpha^6\right)}{1-\alpha}\right] = \frac{-i\left(\alpha-\alpha^7\right)}{1-\alpha}, \quad \left[\because \alpha^7 = e^{i.2\pi} = \cos 2\pi + i\sin 2\pi = 1\right]$
$= \frac{-i(\alpha-1)}{1-\alpha} = i$

Q.24 Answer:

—

Solution:

$\left(x^4+x^2+1\right) = \left(x^2+x+1\right)\left(x^2-x+1\right) = 0$
$\Rightarrow \quad x_1=\omega, x_2=\omega^2, x_3=-\omega, x_4=-\omega^2 \Rightarrow x_1^6+x_2^6+x_3^6+x_4^6=4$

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Q.25 Answer:

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Solution:

$\frac{\sqrt{5+12i}+\sqrt{5-12i}}{\sqrt{(5+12i)}-\sqrt{5-12i}}\ \frac{\sqrt{5+12i}+\sqrt{5-12i}}{\sqrt{(5+12i)}+\sqrt{5-12i}}$
$= \frac{5+12i+5-12i+2\sqrt{5+12i}\sqrt{5-12i}}{5+12i-5+12i} = \frac{10+2\times(\pm 13)}{24i} = -\frac{3}{2}i$ **or** $\frac{2i}{3}$

Q.26 Answer:

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Solution:

We have. $\omega = \left(\frac{z-i}{1+iz}\right)^n = \left(\frac{z-i}{i(z-i)}\right)^n = \left(\frac{1}{i}\right)^n = (-i)^n$
$\therefore \quad |\omega| = |(-i)^n| = |-i|^n = 1$ **for all** η
$\therefore \quad \omega$ **lies on the unit circle for all** n**.**

Q.27 Answer:

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Solution:

$2(\omega+1)\left(\omega^2+1\right)+3(2\omega+1)\left(2\omega^2+1\right)+\ldots\ldots+(n+1)(n\omega+1)\left(n\omega^2+1\right)$
$=\sum_{r=1}^{n}(r+1)(r\omega+1)\left(r\omega^2+1\right)=\sum_{r=1}^{n}(r+1)\left(r^2\omega^3+r\omega+r\omega^2+1\right)$
$\sum_{r=1}^{n}(r+1)\left(r^2-r+1\right)=\sum_{r=1}^{n}\left(r^3-r^2+r+r^2-r+1\right)$
$=\sum_{r=1}^{n}\left(r^3\right)+\sum_{r=1}^{n}(1)=\left[\frac{n(n+1)}{2}\right]^2+n$

Q.28 Answer:

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Solution:

As $z_1, z_2, \ldots\ldots, z_n$ **lie on the circle** $|z|=2, |z_i|=2$
$\Rightarrow \quad |z_i|^2=4 \quad \Rightarrow \quad z_i\bar{z}_i=4$ **for** $i=1,2,3,\ldots,n$
$\frac{1}{z_i}=\frac{1}{4}\bar{z}_i$
for $i=1,2,3,\ldots,n$
$\therefore E=|z_1+z_2+\ldots..+z_n|-4\left|\frac{1}{4}\bar{z}_1+\frac{1}{4}\bar{z}_2+\ldots\ldots+\frac{1}{4}\bar{z}_n\right|$
$=|z_1+z_2+\ldots\ldots+z_n|-|\bar{z}_1+\bar{z}_2+\ldots\ldots+\bar{z}_n|$
$=|z_1+z_2+\ldots\ldots+z_n|-|\overline{z_1+z_2+\ldots\ldots+z_n}|=0 \quad \therefore \quad |z|=|\bar{z}|]$

Q.29 Answer:

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Solution:

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$x^2 - 2x\cos\theta + 1 = 0 \Rightarrow x = \cos\theta \pm i\sin\theta$

$\Rightarrow \quad x^{2n} = \cos 2n\theta \pm i\sin 2n\theta$ **and** $x^n = \cos n\theta \pm i\sin n\theta$

$\therefore \quad x^{2n} - 2x^n\cos n\theta + 1 = \cos 2n\theta \pm i\sin 2n\theta - 2\cos^2 n\theta \square 2i\sin n\theta\cos n\theta + 1 = 0$

Q.30 Answer:

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Solution:

As we know that $\mathrm{i} + \mathrm{i}^2 + \mathrm{i}^3 + \mathrm{i}^4 = 0$

$\sum_{n=1}^{13}\left(i^n + i^{n+1}\right) = \sum_{n=1}^{13}(1+i)\left(i^n\right) = (1+i)\sum_{n=1}^{13} i^n = (1+i)\left(0 + 0 + i^{13}\right) = (1+i)i = i + i^2 =$**i-1**

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